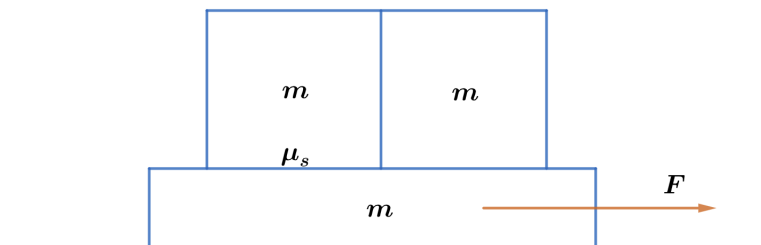


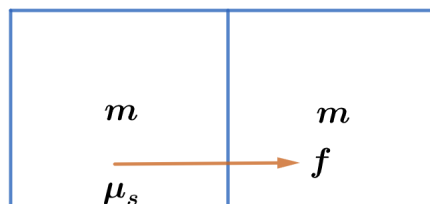
2023 F=ma Exam: Problem 7

Kevin S. Huang



If the upper boxes don't slide, then all three boxes move together. Applying Newton's 2nd law to the combined system of boxes, we have

$$F = 3ma$$
$$a = \frac{F}{3m}$$



Applying Newton's 2nd law to the upper two boxes, we have

$$f = 2ma$$

We know the maximum value of static friction $f \leq \mu_s N = \mu_s mg$ so

$$2ma \leq \mu_s mg$$
$$\mu_s \geq \frac{2a}{g} = \frac{2F}{3mg}$$

Thus, the answer is D.